Case Study No.: 4

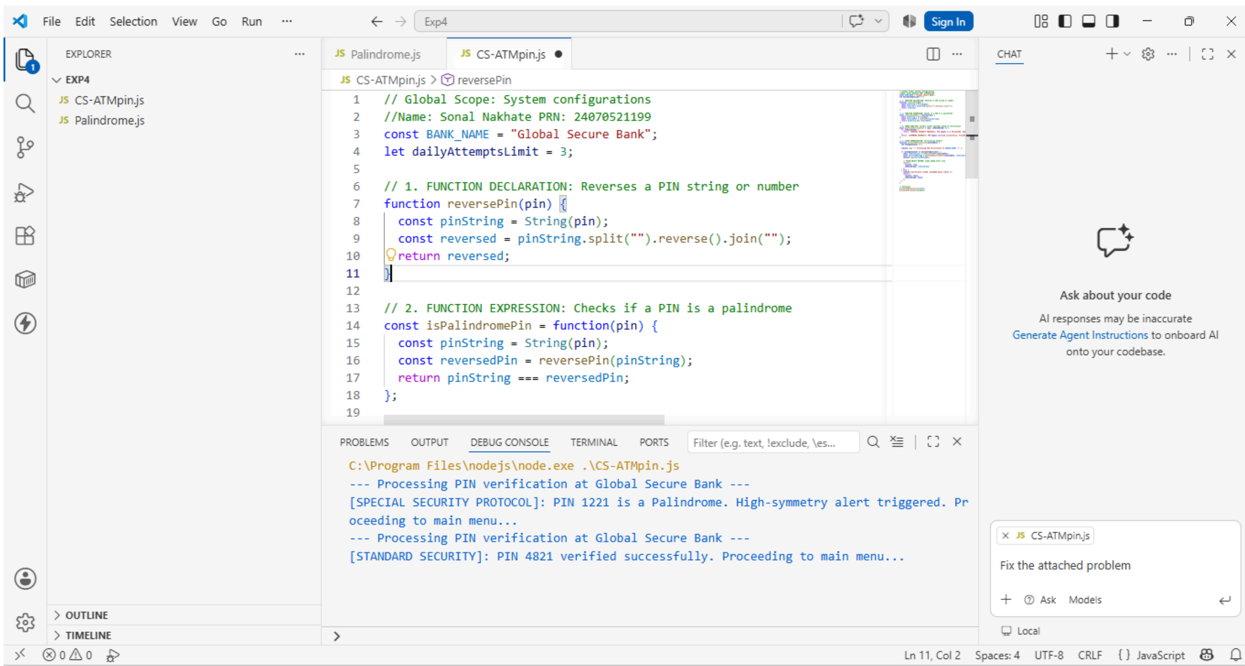
1. Case Study Title: ATM Card Pin Verification using JavaScript Functions Case Study.

2. Case Study Program Code:

```
// Global Scope: System configurations
//Name: Sonal Nakhate PRN: 24070521199
const BANK_NAME = "Global Secure Bank";
let dailyAttemptsLimit = 3;

// 1. FUNCTION DECLARATION: Reverses a PIN string or number
function reversePin(pin) {
  const pinString = String(pin);
  const reversed = pinString.split("").reverse().join("");
  return reversed;
}

// 2. FUNCTION EXPRESSION: Checks if a PIN is a palindrome
const isPalindromePin = function(pin) {
  const pinString = String(pin);
  const reversedPin = reversePin(pinString);
  return pinString === reversedPin;
};

// 3. ARROW FUNCTION: Formats output messages based on verification
const generateSecurityAlert = (pin, isPalindrome) => {
  if (isPalindrome) {
    return `[SPECIAL SECURITY PROTOCOL]: PIN ${pin} is a Palindrome.
High-symmetry alert triggered. Proceeding to main menu...`;
  }
```

```
  return `[STANDARD SECURITY]: PIN ${pin} verified successfully.
Proceeding to main menu...`;
};

// 4. SCOPE DEMONSTRATION: Verification Handler
function processATMTransaction(enteredPin) {
  let attemptCounter = 1;

  console.log(`--- Processing PIN verification at ${BANK_NAME} ---`);

  if (attemptCounter <= dailyAttemptsLimit) {
    const isPalindromic = isPalindromePin(enteredPin);
    const securityMessage = generateSecurityAlert(enteredPin,
isPalindromic);
    console.log(securityMessage);

    // FIXED OBJECT RETURN: Comma added after true
    return {
      success: true,
      isPalindrome: isPalindromic
    };
  } else {
    console.log("Account locked. Exceeded daily limits.");
    return {
      success: false,
      isPalindrome: false
    };
  }
}

// Execution
processATMTransaction(1221);
processATMTransaction(4821);
```

3. Output:



4. Result/Conclusion: We have successfully implemented a simple JavaScript program that checks if the ATM pin is a palindrom or not.